

CET138 HTML: HyperText Markup Language

Introduction

HTML (**HyperText Markup Language**) is the foundation of every website. It is a markup language used to structure content on the web using a system of tags and elements. HTML provides semantic meaning to web content, telling browsers how to display text, images, links, forms, and other media.

HTML is not a programming language. It is declarative metadata that describes the structure and meaning of content. Key concepts include:

- Tags and Elements: Used to mark up content (e.g., h1, p, img)
- Attributes: Provide additional information about elements (e.g., href, src, class)
- Semantic HTML: Using meaningful tags that describe content purpose (e.g., header, nav, article)
- Form Elements: Collecting user input through input, textarea, select, and button

Code Snippet index.html Head & Hero Section

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Rajesh Karki</title>
  <link href="bootstrap.min.css" rel="stylesheet"/>
  <link href="css/style.css" rel="stylesheet"/>
</head>
<body>
  <h1 class="hero-h1">
    Building the
    <span class="glow" id="typedText">Frontend Magic</span>
    of the Web
  </h1>
  <p class="hero-desc">
    Welcome to my CET138 Full Stack Development ePortfolio.
  </p>
  <div class="hero-cta d-flex flex-wrap gap-3">
    <a href="about.html" class="btn-primary-custom">Explore Portfolio</a>
    <a href="contact.html" class="btn-outline-custom">Get in Touch</a>
  </div>
</body>
</html>
```

Explanation of Code

- <!DOCTYPE html>: Declares that this document is HTML5. Every HTML document must begin with this.
- <html lang="en">: Root element; the lang attribute specifies the language as English.
- <meta charset="UTF-8"/>: Specifies character encoding for proper display of special characters.
- <meta name="viewport" .../>: Essential for responsive design matches page width to device width.
- <title>: The text that appears in the browser tab.

- `<link>` tags: Link External Resources Like Bootstrap and the custom stylesheet.
- `<h1>`, `<p>`: Semantic heading and paragraph tags for content hierarchy.
- `class` attribute: Applies CSS styles and JavaScript functionality to elements.
- `<a>` tags: Anchor/hyperlink elements for navigating between pages.

Demonstration of Understanding

- Proper Document Structure: Correct DOCTYPE, `html`, `head`, and `body` tags.
- Semantic Elements: Appropriate heading levels and paragraph tags for hierarchy.
- Meta Tags: Essential meta tags for character encoding and responsive design.
- Resource Linking: Properly links external CSS stylesheets and frameworks.
- Accessibility: Semantic HTML supports screen readers and improves accessibility.
- Responsive Design: Includes the `viewport` meta tag needed for mobile responsiveness.

Conclusion

HTML is the starting point for all web development. While HTML alone creates plain text and structure, it is the essential foundation upon which CSS styling and JavaScript interactivity are built. Understanding how to write semantic, properly structured HTML is crucial for creating accessible, maintainable, and SEO-friendly web applications.